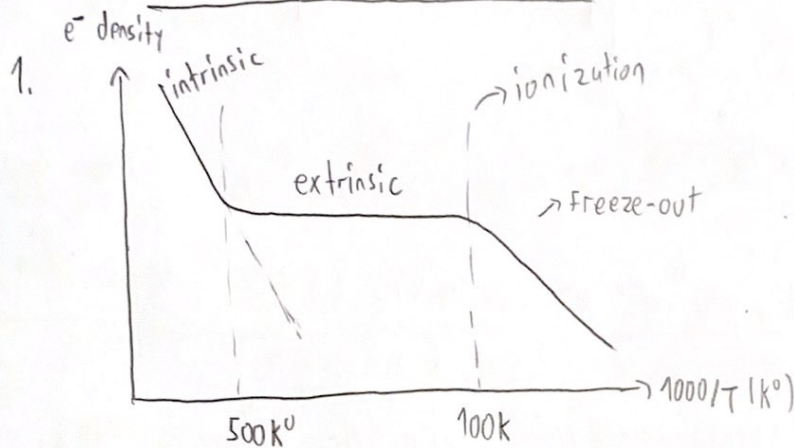
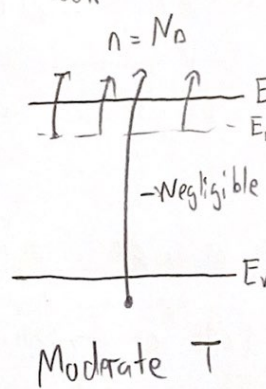
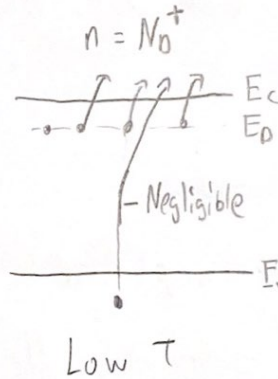


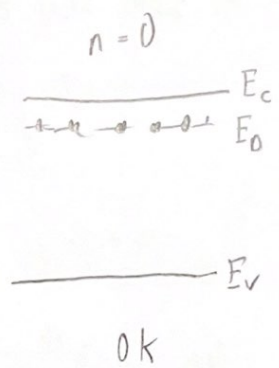
Lots of electrons jumped to conduction band from valence band. Intrinsic electrons dominates the number of dopant electrons.



(Almost) all of the electrons are ionized. Some jumps from the valence band also occurs but it's negligible compared to N_D . So $n = N_D$



As heat increases electrons start to ionize and jump to conduction band. Not all (but some) of the electrons are ionized. Also there are few e^- 's jumping from the valence band.



None of the electrons are ionized. They're all in donor level

2. At 300K (room temp) it's known that

$$n_i = p_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$n_i(T) = 2 \left(\frac{2\pi kT}{h^2} \right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

$m_n^* = 1.18 m_0$, $m_p^* = 0.81 m_0$, $m_0 = 9.11 \times 10^{-31}$, $E_g = 1.12 \text{ eV}$, $k = 1.38 \times 10^{-23} \text{ J/K}$, $h = 6.63 \times 10^{-34} \text{ J.s}$

Also $125^\circ = 398K \Rightarrow kT = 0.024 \text{ eV} \frac{398}{300} = 0.034 \text{ eV}$

$$2. \quad n_i(398K) = 2 \left[\frac{2\pi \times 1.38 \times 10^{-23} \times 398}{(6.63 \times 10^{-34})^2} \right]^{3/2} \left[1.18 \times 0.81 \times (9.11 \times 10^{-31})^2 \right]^{3/4} e^{-\frac{1.12 \text{ eV}}{2 \times 0.026 \text{ eV}}}$$

$$n_i(398K) = 2.6 \times 10^{12} \text{ cm}^{-3}$$

At room temp since $N_D = 10^{16} \text{ cm}^{-3} \gg n_i(300K) = 1.5 \times 10^{10} \Rightarrow n_0 = 10^{16} \text{ cm}^{-3}$

At 398K still $N_D = 10^{16} \text{ cm}^{-3} \gg n_i(398K) = 2.6 \times 10^{12} \text{ cm}^{-3} \Rightarrow$

At 398K
 $n_0 = N_D = 10^{16} \text{ cm}^{-3}$ still

3.

Because of the charge compensation principle $n_0 = N_D - N_A$

$N_D = 4 \times 10^{16} \text{ cm}^{-3} \rightarrow \text{Arsenic}$, $N_A = 1 \times 10^{16} \text{ cm}^{-3} \rightarrow \text{Boron}$

$$n_0 = 3 \times 10^{16} \text{ cm}^{-3}, \quad n_0 p_0 = n_i^2 \Rightarrow p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{3 \times 10^{16}} = 7.5 \times 10^3 \text{ cm}^{-3}$$

$$n_0 = n_i e^{-(E_F - E_i)/kT}$$

$$\Rightarrow E_F - E_i = kT \ln\left(\frac{n_0}{n_i}\right)$$

$$E_F - E_i = 0.026 \text{ eV} \times \ln\left(\frac{3 \times 10^{16}}{1.5 \times 10^{10}}\right) = 0.38 \text{ eV}$$

$$4. \quad f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

$$E - E_F = 0.1 \text{ eV}$$

a) $E - E_F = 0.1 \text{ eV}$ $T = 0 \text{ K}$ $f(E) = \frac{1}{1 + e^{0.1/0}} = \frac{1}{\infty} = 0$

b) $T = 300 \text{ K}$ $f(E) = \frac{1}{1 + e^{0.1/0.026}} = 0.02$

c) $T = 500 \text{ K}$ $kT = 0.026 \times \frac{500}{300} = 0.043 \text{ eV}$ $f(E) = \frac{1}{1 + e^{0.1/0.043}} = 0.009$

5. We can find from the tables that mobility of an electron at 300 K with a doping concentration of 10^{14} cm^{-3} is $461 \text{ cm}^2/\text{V}\cdot\text{s}$

$$n_0 = 10^{14} \text{ cm}^{-3} \quad n_i = 1.5 \times 10^{10} \text{ cm}^{-3} \Rightarrow p_0 = \frac{n_i^2}{n_0} = 2.25 \times 10^6 \ll n_0$$

p_0 is negligible.

$$\rho = \frac{1}{\sigma} = \frac{1}{\sigma_n + \sigma_p} = \frac{1}{qn\mu_n + qp\mu_p} \approx \frac{1}{qn\mu_n}$$

resistivity

$$\rho = \frac{1}{1.6 \times 10^{-19} \text{ C} \times 10^{14} \text{ cm}^{-3} \times 461 \text{ cm}^2/\text{V}\cdot\text{s}} = \boxed{136 \text{ } \Omega\cdot\text{cm}}$$

6. $\frac{D}{\mu} = \frac{kT}{q}$ $kT = 26 \text{ meV}$ at 300 K, $\mu_n = 1250 \text{ cm}^2/\text{V}\cdot\text{s}$

$$D = \mu \frac{kT}{q} = 1250 \text{ cm}^2/\text{V}\cdot\text{s} \times \frac{0.026 \text{ eV}}{1 \text{ e}} = 1250 \times 0.026 \frac{\text{cm}^2}{\text{V}\cdot\text{s}} = \boxed{32.5 \text{ cm}^2/\text{s}}$$

→ 7 and 8th were instructed to skip by Şenol Mutlu as we haven't covered these topics yet.